

TOI-1811b

R-2362

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI1811_240112 · created 2026-08-19T23:34:36+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460321.9605 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1481 +/- 0.007
Transit depth (Rp/Rs)^2	2.19 +/- 0.21 [%]
Orbital Inclination (inc)	89.73 +/- 0.91
Airmass coefficient 1 (a1)	0.8683 +/- 0.0048
Airmass coefficient 2 (a2)	0.142 +/- 0.048
Scatter in the residuals of the lightcurve fit is	0.55 %
Best Comparison Star	#3 - [118, 213]
Optimal Aperture	5.43
Optimal Annulus	15.4
Transit Duration (day)	0.087 +/- 0.0044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1481 ± 0.007 vs 0.13272 (deviation 2.2σ)
Duration fit vs published	0.087 d vs 0.0813 d (deviation 1.31σ)
Mid-time O–C	-22.2 min
Depth SNR	21.2
Residual scatter	0.55 %

Light curve

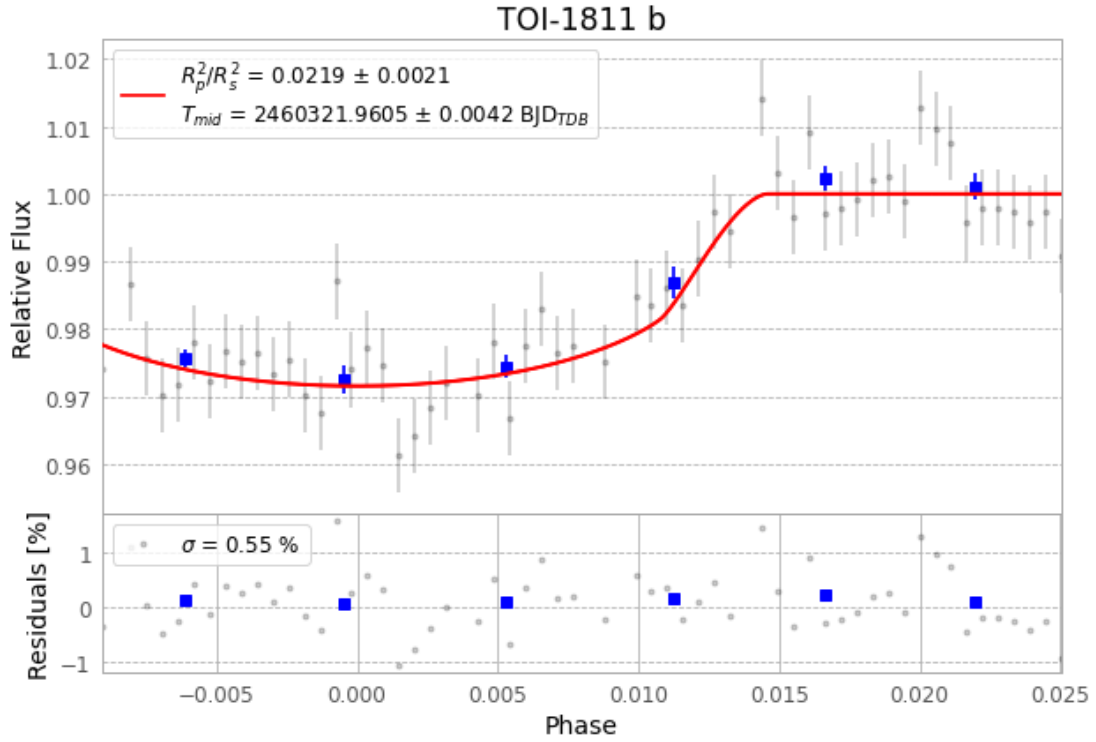


Figure 1 — FinalLightCurve_TOI-1811 b_2024-01-12.png (SHA-256 e7084feeb02eef08...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [401, 271], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8969ddb1ca808104...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

61 FITS frames (40 MB), TOI1811240112100946.FITS → TOI1811240112131216.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI1811-240112.log (fdfe61b92365...), FinalLightCurve_TOI-1811 b_2024-01-12.png (e7084feeb02e...), FinalLightCurve_TOI-1811 b_2024-01-12.csv (07366705d34b...)

Compute resources

Wall time 452.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 23:34Z.